

Mankiewicz extension on open connected domains

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension`

Glues local affine-isometry charts to obtain the principal open-domain extension theorem.

Statement

Let f be a surjective isometry from an open connected subset s of a real normed space onto an open subset t of another real normed space. Then there exists a unique ambient real affine isometry equivalence A agreeing with f on all of s . Target connectedness is not a separate assumption.

Assumptions

Real normed vector spaces E and F ; s is open and connected; t is open; f is a surjective isometry from s to t .

Conclusion

There is a unique ambient real affine isometry equivalence whose restriction to s is f .

Proof idea

At each source point, openness of s and t supplies matching small ambient balls. The local chart theorem gives an affine isometry agreeing with f on a smaller ball around that point. These local charts agree on overlaps and therefore form a locally constant chart-valued map; connectedness of s makes it constant. The resulting global chart is the unique extension.

Proof steps

1. For each $x \in s$, choose positive source and target ball radii from openness and take their minimum R .
2. Apply the local affine-chart theorem to obtain an ambient affine isometry agreeing with f on $\text{ball}(x, R/8)$.
3. Feed these pointwise local charts into the connected gluing lemma.
4. The gluing lemma returns one affine isometry agreeing with f everywhere on s and proves uniqueness from agreement on the nonempty open source.

Main citations

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.OpenConnected.0.MathlibAnnex.IsometryEquiv.existsUnique_affineIsometryEquiv_of_local`

The proof or construction of Mankiewicz extension on open connected domains uses the project declaration “Gluing locally defined affine isometries on a connected open set” at the indicated step.

`MathlibAnnex.IsometryEquiv.exists_affineExtension_eqOn_ball`

The proof or construction of Mankiewicz extension on open connected domains uses the project declaration “Local affine-isometry chart on a smaller ball” at the indicated step.

Lean source signature (exact)

```
theorem existsUnique_affineExtension
  {s : Set E} {t : Set F} (f : s  $\approx_i$  t)
  (hs : IsOpen s) (hsc : IsConnected s) (ht : IsOpen t) :
   $\exists!$  A : E  $\approx^{a.i}[\mathbb{R}]$  F,
   $\forall$  x : s, A (x : E) = ((f x : t) : F)
```

[Read exact MathlibAnnex v0.4.0 source](#)

Exact Card identity

Stable Card ID: 30b5b915102151510645ff9e55aef46c99b259d2cb09540049f2b73b0eb07373

Card SHA-256: 1263484c712e461a6802349daed7936b3d5bc3f619a4670a7d5bc843f8a3263f

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